Connected Component Analysis for Multi-Vehicle Detection in Shipment Tracking Systems

B.Tech Project Presentation

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Problem & Motivation

The global RTTVP market is valued at ≈USD 3.2 billion. Leading platforms process over **8 billion data points daily** across 1.5 billion annual shipments.

A critical data quality failure: a single shipment tracked by **multiple vehicles simultaneously**, producing position histories no single physical object could have generated.

Data quality dimensions at risk:

- **Accuracy:** GPS coordinates vs. actual location
- **Completeness:** gaps from tunnels, battery loss
- **Consistency:** physically plausible trajectories
- **Validity:** correct formats and value ranges

Business Impact

- Customer confusion & erratic tracking pages
- Incorrect ETAs & dock-door scheduling
- Billing disputes (phantom mileage)
- Operational blindness
- SLA violations & expedited freight costs

Poor data quality costs organisations an average of $12.9M annually.

Error Scenarios: How Multi-Vehicle Binding Occurs

Scenario A: Device Misconfiguration

Multiple GPS devices bound to same shipment ID (wrong IMEI mapping, failed device swap).

Scenario B: Data Integration Errors

Shipment ID collisions when merging carrier feeds (e.g. Carrier A and Carrier B both use `SHIP-12345`). No carrier-scoped namespaces.

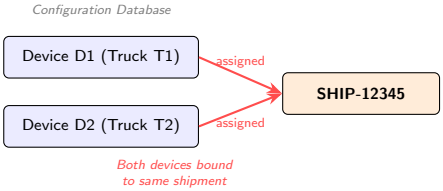
Scenario C: Duplicate Transmissions

Malfunctioning device or network issues create duplicate position records with slightly different coordinates.

Scenario D: System Migration Issues

Historical data from old devices incorrectly linked with new device data during platform migration.

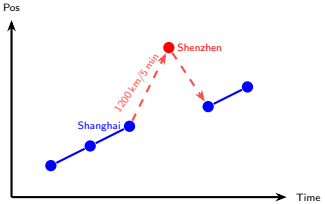
Scenario A Illustration



Result

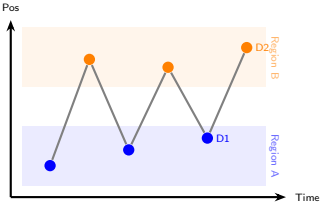
Dashboard shows shipment alternating between two corridors. Hanover (D1) and Leipzig (D2), 250 km apart in 3 s.

Impossible Teleportation



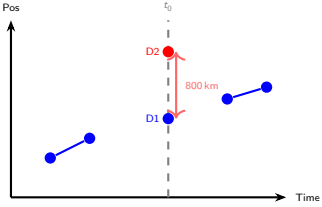
Sudden jump: 14,400 km/h implied speed

Oscillating Locations



Interleaved pings from 2 devices

Concurrent Distant Positions

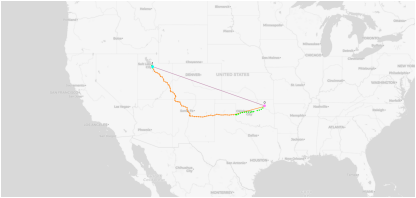


Same timestamp, 800 km apart

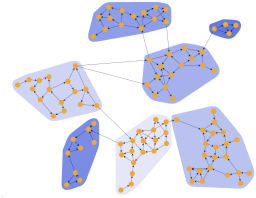
Challenge

Must distinguish these anomalous patterns from **legitimate handovers**: truck → ship at port, inter-modal transfer at hub, relay handoff on highway. Key difference: legitimate handovers have a *spatial and temporal bridge* (common meeting point within feasible time).

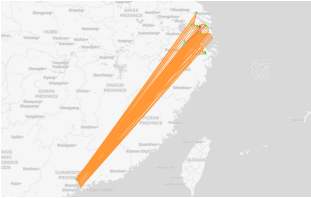
The Problem in Practice



A typical GPS-based shipment tracking interface



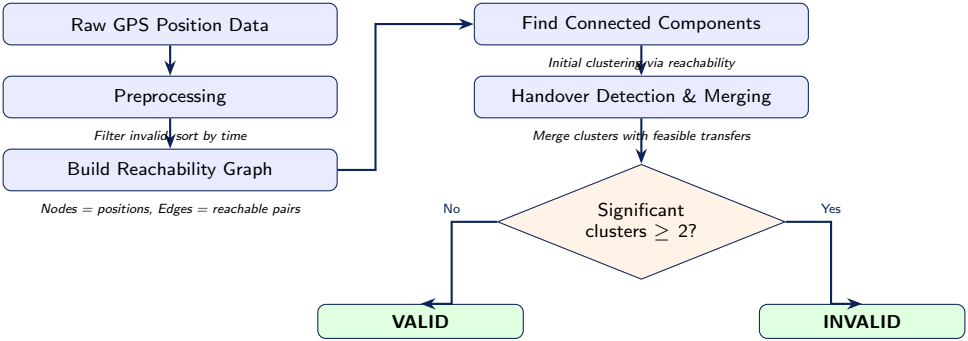
Connected components color-coded by cluster



Before: Conflicting GPS trails from two devices

After: Trails separated, anomaly identified

Proposed Framework: System Overview



Graph Model & Formulation

Notation

$p_i = (\text{lat}, \text{lon}, t)$	Position
$P = \{p_1, \ldots, p_n\}$	Sorted sequence
$V_{\max} = 250\,\text{km/h}$	Max feasible speed
$\tau = 1800\,\text{s}$	Jitter window
$T_H = 28800\,\text{s}$	Handover window
$\alpha = 0.10$	Min cluster %
$s_{\min} = 5$	Min cluster size

Reachability Edge

Edge (p_i, p_j) exists iff:

$$\underbrace{d(p_i, p_j)}_{\text{Haversine}} \leq V_{\max} \cdot (t_j - t_i)$$

Nodes = GPS positions. Graph is always a **DAG** (time-ordered, no cycles).

Jitter Tolerance

GPS noise handled by checking **multiple recent** positions in window τ:
$\exists\, p_i \in W(p_j, C, \tau) : \text{reachable}(p_i, p_j)$
W: positions in cluster C within $\tau = 30\,\text{min}$ before p_j.

Handover Detection

Legitimate inter-vehicle transfers detected via **boundary positions**:
$\exists\, p_i \in \text{last}_k(C_1),\ p_j \in \text{first}_k(C_2) : |t_i - t_j| \leq T_H \ \wedge\ \text{reachable}(p_i, p_j)$
$k = 5$ boundary positions, $T_H = 8\,\text{h}$.

Significance Filter

A cluster is **significant** iff $|C| \geq \max(s_{\min},\ \alpha \cdot n)$.
Filters GPS jitter / outlier clusters from the multi-vehicle decision.

Key Theorems & Correctness

Theorem 1: Single Component $\Rightarrow$ Single Vehicle

One weakly connected component $\Rightarrow$ all positions consistent with one vehicle.

Proof: Every position reachable from every other $\Rightarrow$ feasible single-vehicle trajectory.

Theorem 2: Multiple Large Components $\Rightarrow$ Multiple Vehicles

G has ≥ 2 components each with $\geq 10\%$ of positions $\Rightarrow$ multiple vehicles.

Proof: Two large components cannot both be noise; no edges between them $\Rightarrow$ separate vehicles.

Theorem 3: Handover Preserves Validity

Two components merging via valid handover $\Rightarrow$ valid single-vehicle tracking with transfer.

Lemma 1: Completeness

All positions from one vehicle $\Rightarrow$ algorithm returns **TRUE**.

Proof: Single vehicle positions are mutually reachable; clustering yields one component $\Rightarrow$ TRUE.

Lemma 2: Soundness

Two vehicles at distant locations $\Rightarrow$ algorithm returns **FALSE**.

Proof: Impossible speeds prevent edges across vehicles $\Rightarrow$ separate significant clusters $\Rightarrow$ FALSE.

Mapping to Standard Problems

Our Problem	Standard Problem
Initial clustering	Connected components
Cluster merging	Iterative pairwise union
Significance test	Component size threshold
Handover detection	Boundary reachability

Algorithm: SingleVehicleCheck

Key Sub-Algorithms

1. PreprocessPositions: filter & sort
2. BuildInitialClusters: online clustering
3. IsReachable: Haversine + speed check
4. GetRecentPositions: jitter window
5. MergeClustersViaHandovers: iterative merge
6. CanHandover: boundary reachability
7. MakeDecision: significance threshold

Clustering (Phase 2): Pseudocode

for each position p chronologically:
  for each existing cluster C (newest first):
    $recent \leftarrow$ last 3 positions of C within τ
    if any $q \in recent$ reachable from p: assign p to C; **break**
  if not assigned: create new cluster $\{p\}$
Merge clusters via handover predicate (iterate until stable).

Decision (Phase 3)

$threshold \leftarrow \max(s_{\min},\ \alpha \cdot n)$ 0 sig. clusters $\rightarrow$ **UNCERTAIN** $\cdot$ 1 $\rightarrow$ **TRUE** $\cdot$ $\geq 2 \rightarrow$ **FALSE**

Phase 1: Preprocessing
Filter nulls/NaN, validate ranges, sort

↓

Phase 2: Clustering
Online reachability + handover merging

↓

Phase 3: Decision
Count significant clusters → result

Haversine Distance

$$d = 2R \cdot \arcsin\sqrt{\sin^2\frac{\Delta\phi}{2} + \cos\phi_i \cos\phi_j \sin^2\frac{\Delta\lambda}{2}}$$

$R = 6{,}371\,\text{km}$. Edge exists iff $d \leq V_{\max} \cdot \Delta t$, $V_{\max} = 250\,\text{km/h}$.

Worked Example: Shanghai & Shenzhen (12 Positions)

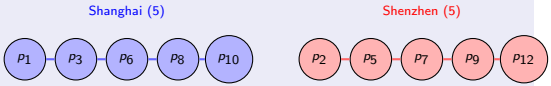
Input: 12 Raw GPS Positions

p_1:	Shanghai	31.23°N	121.47°E	08:00
p_2:	Shenzhen	22.55°N	114.05°E	08:15
p_3:	Shanghai	31.24°N	121.46°E	09:00
p_4:	NaN			09:30
p_5:	Shenzhen	22.56°N	114.06°E	10:00
p_6:	Shanghai	31.25°N	121.48°E	10:30
p_7:	Shenzhen	22.54°N	114.07°E	11:00
p_8:	Shanghai	31.22°N	121.49°E	11:30
p_9:	Shenzhen	22.57°N	114.04°E	12:00
p_{10}:	Shanghai	31.26°N	121.45°E	12:30
p_{11}:	NaN			13:00
p_{12}:	Shenzhen	22.58°N	114.08°E	13:30

Step 1 – Preprocess: Remove p_4, p_{11} (NaN) $\rightarrow$ 10 valid positions.

Step 2 – Cluster: Shanghai–Shenzhen $\approx$ 1,200 km. Transitions require $>$ 4,800 km/h. No cross-cluster edges.

Step 3: Connected Components



Step 4: Handover Check

Boundary positions of clusters are $\approx$ 1,200 km apart. No feasible handover $\Rightarrow$ no merging.

Step 5: Decision

$n = 10$, threshold $= \max(5,\ 0.1 \times 10) = 5$
Cluster sizes: 5 (Shanghai) ≥ 5 ✓, 5 (Shenzhen) ≥ 5 ✓
Significant clusters $= 2 \Rightarrow$ **Result: FALSE** (Multiple vehicles detected).

Complexity Analysis & Edge Cases

Phase	Time	Space
Preprocessing	$O(n \log n)$	$O(n)$
Initial Clustering	$O(n \cdot m)$	$O(n)$
Handover Merge	$O(m^3 \cdot b^2)$	$O(m)$
Decision	$O(m)$	$O(1)$
Total	$O(n \log n + m^3 b^2)$	$O(n)$

$n =$ positions, $m =$ clusters ($m \ll n$), $b = 5$.
Typical case ($m < 10$): effectively $O(n \log n)$.

Edge Case Handling

Case	Result
No positions	UNCERTAIN
Single position	TRUE
All invalid	UNCERTAIN
$\Delta t = 0$	100 m tolerance

Input Validation

- Null & NaN checks on coordinates
- Range: lat $\in [-90, 90]$, lon $\in [-180, 180]$
- Timestamp validation
- Empty input $\rightarrow$ UNCERTAIN

Thank You

Questions?